MTH401 - Differential Equations (Quiz # 3)

Question # 5 of 10 (Start time: 01:26:30 PM, 26 June 2025)

The total forces acting on mass m are_____.

Select the correct option

◉	2
○	3
○	4
○	5

MTH401 - Differential Equations (Quiz # 3)

Question # 7 of 10 (Start time: 01:27:24 PM, 26 June 2025)

How are frequency and period related in simple harmonic motion?

Select the correct option

○	None of the above
●	They are inversely related
○	Their sum is constant
○	They are directly related

MTH401 - Differential Equations (Quiz # 3)

Question # 1 of 10 (Start time: 01:24:38 PM, 26 June 2025)

If the auxiliary equation $m^3 + m = 0$ has roots $m = 0, m = \pm i$ then the complementary function is

Select the correct option

○	$y_c = c_1 \cos x + c_2 \sin x$
◉	$y_c = c_1 + c_2 \cos x + c_3 \sin x$

MTH401 - Differential Equations (Quiz # 3)

Question # 9 of 10 (Start time: 01:27:49 PM, 26 June 2025)

If

$$\lambda^2 - \omega^2 = 0$$

and

$$\beta = k$$

then the system is said to be _____ damped.

Select the correct option

○	under
◉	critically
○	none of these
○	over

MTH401 - Differential Equations (Quiz # 3)

Question # 8 of 10 (Start time: 01:27:36 PM, 26 June 2025)

Consider a power series

$$1 - \frac{x^2}{2} + \frac{x^4}{24} - \ldots.$$

represents _____.

Select the correct option

○	sin x
○	e
○	ln x
◉	cos x

MTH401 - Differential Equations (Quiz # 3)

Question # 4 of 10 (Start time: 01:26:16 PM, 26 June 2025)

The damping force is _____to the instantaneous velocity

$$\frac{dx}{dt}$$

.

Select the correct option

○	Constant
○	None of these
○	Inverse proportional
◉	Proportional

MTH401 - Differential Equations (Quiz # 3)

Question # 6 of 10 (Start time: 01:27:09 PM, 26 June 2025)

If the system is impressed upon by a _____ force and there is no damping force then there is no transient term in the solution.

Select the correct option

○	gravitational
○	applied
○	friction
◉	periodic

Click to

MTH401 - Differential Equations (Quiz # 3)

Question # 10 of 10 (Start time: 01:28:02 PM, 26 June 2025)

For a second order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = 2$, $W_1 = \dfrac{e^{-x}}{x}$, then $u'_1 =$ _____.

Select the correct option

○	$\dfrac{e^{-x}}{2x}$
○	$2x\,e^{-x}$
○	$\dfrac{e^{-x}}{x}$
●	$-\dfrac{e^{-x}}{2x}$

MTH401 - Differential Equations (Quiz # 3)

Quiz Start Time: 01:24 PM, 26 June 2025

Question # 2 of 10 (Start time: 01:25:06 PM, 26 June 2025)

Total Marks: 1

If a force acts upon a body, the acceleration is produced in the direction of the force whose magnitude is proportional to the magnitude of force, is known as the _____.

Select the correct option

●	Newton's second law
○	Newton's first law
○	Hook's law
○	Newton's third law

Click to Save Answer & Move to Next Question

MTH401 - Differential Equations (Quiz # 3)

Question # 3 of 10 (Start time: 01:25:20 PM, 26 June 2025)

A power series in (x-2) is an infinite series of the form

$$\sum_{n=0}^{\infty} c_n(x-2)^n = c_0 + c_1(x-2) + c_2(x-2)^2 + \dots.$$

the number 2 is called _____.

Select the correct option

○	Radius of power series
○	Base of power series
○	none of these
◉	Center of power series